OS Issues and Tools

Objectives:

At the end of this episode, I will be able to:

1. Identify common operating system issues.

Exam Objective: *4.4 - Given a scenario, troubleshoot common OS and software problems.*

Description: In this episode, the viewer will identify software tools used to troubleshoot, diagnose and repair common operating system and software issues such as patch management, recovery solutions, administrative accounts, reboots, service and process management as well as configuration management.

- Common problems
  - Unable to log on - Domain (Clock skew)/Account issues
  - Unable to access resources - Domain/permissions, application-level permissions
  - Unable to access files - Domain/permissions, application-level permissions
  - System file corruption - Malware, update failures, device failure
  - End of life/end of support - lack of patching
  - Slow performance - Workload/available resoures, disk space, fragmentation, memory leaks
  - Cannot write to system logs - application-level permissions
  - Service failures - Disabled, start failure, dependency issues
  - System or application hanging - compatibility issues, file corruption/missing, memory leaks, buffer overrun
  - Freezing - incompatibility issues, file corruption/missing, drivers, modules, memory leaks
  - Patch update failure - network connection, incompatibility, disk space, application-level
- OS and software tools and techniques
  - Patching
    - Upgrades
    - Downgrades
    - Schedule reboots
  - Package management
  - Proper privilege escalations
    - runas/Run As
    - sudo
    - su
  - Software firewalls
    - Adding or removing ports
    - Zones
  - Clocks
    - Network time protocol (NTP)
    - System time
  - Services and processes
    - Starting
    - Stopping
    - Status identification
    - Dependencies
  - Configuration management
    - System center configuration manager (SCCM)

        - Puppet/Chef/Ansible
        - Group Policy Object (GPO)
    - Hardware compatibility list (HCL)
    - Recovery
        - Boot options
            - Safe mode
            - Single user mode
        - Reload OS
        - Snapshots

External Resources:

During this episode, you can reference the following external resources for supplementary tools and information:

- Not applicable if blank